ASPIRIN

Not for children <16 yr

Bleeding risk

CHEMICAL NAME 2-(acetyloxy)benzoic acid (acetylsalicylic acid)
CLASS Antiplatelet (irreversible COX inhibitor); salicylate NSAID
TRADE NAMES Aspirin Protect · Cardioaspirin · Aspocid · Ecotrin

THERAPEUTIC USES / INDICATIONS

- Acute coronary syndrome (300 mg chewed immediately) and secondary prevention of MI/stroke.
- Acute ischaemic stroke (after haemorrhage excluded) and TIA.
- Pre-eclampsia prevention (low dose from 12 weeks in high-risk women).
- After coronary stents (with P2Y12 inhibitor); Kawasaki disease (specialist); pain/fever (higher doses, rarely now).

ADMINISTRATION

- **PO** with or after food; enteric-coated tablets swallowed whole.
- In ACS: 300 mg non-enteric-coated, chewed for rapid effect.
- Dispersible tablets dissolved in water.

DOSAGES

- **ACS loading:** 300 mg chewed, then 75–100 mg daily.
- **Secondary prevention:** 75–100 mg once daily.
- **Acute ischaemic stroke:** 300 mg daily for 2 weeks, then 75 mg (or clopidogrel).
- **Pre-eclampsia prevention:** 75–150 mg daily from 12 weeks until birth (high/moderate risk).
- **Analgesia:** 300–900 mg every 4–6 h (max 4 g/day).

PHARMACOKINETICS

- **Onset:** antiplatelet within 1 h (chewed 15–20 min). **Duration:** effect lasts platelet lifetime (7–10 days).
- Rapidly hydrolysed to salicylate in gut wall/liver.
- Excreted in urine (alkaline urine ↑ excretion). **t½:** aspirin 15–20 min; salicylate 2–3 h (dose-dependent).

PHARMACODYNAMICS

- Irreversibly acetylates COX-1 in platelets → ↓ thromboxane A_2 → ↓ platelet aggregation.
- Higher doses inhibit COX-2 → analgesic, antipyretic, anti-inflammatory.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dyspepsia, nausea.
- Easy bruising.
- Minor bleeding (nosebleeds, gums).

ADVERSE EFFECTS (SERIOUS)

- GI ulceration and bleeding.
- Haemorrhagic stroke (small ↑).
- Bronchospasm in aspirin-exacerbated respiratory disease.
- Reye's syndrome in children; salicylate toxicity (tinnitus, hyperventilation, metabolic acidosis).

CONTRAINDICATIONS & CAUTIONS

- Children <16 yr (Reye's syndrome) — except specialist indications (Kawasaki).
- Active peptic ulcer; haemophilia and bleeding disorders.
- Aspirin/NSAID hypersensitivity (asthma, urticaria).
- Third trimester at analgesic doses; severe hepatic/renal impairment.

KEY INTERACTIONS

- **Anticoagulants, other antiplatelets, SSRIs, steroids:** ↑ bleeding.
- **Ibuprofen:** may block cardioprotective effect (give aspirin first).
- **Methotrexate:** ↑ toxicity.
- **Alcohol:** ↑ GI bleeding.

NURSING MANAGEMENT

- Check allergy/asthma and bleeding history.
- In chest pain: give 300 mg chewed promptly unless contraindicated.
- Monitor for GI bleeding (black stools, haematemesis), bruising.
- Consider PPI co-prescription in high-risk patients.
- Hold 5–7 days before surgery only if instructed by surgeon/cardiologist.

PATIENT EDUCATION

- Take with food; do not stop without medical advice (stent thrombosis risk).
- Report black stools, vomiting blood, unusual bruising.
- Avoid other NSAIDs; limit alcohol.
- Do not give to children for fever.

KEY POINTS

- Irreversible COX-1 inhibitor; antiplatelet.
- No aspirin in children <16 (Reye's).
- ACS: 300 mg chewed.
- GI bleeding; effect lasts 7–10 days.

SOURCES CHECKED **[404]** Aspirin 300 (UK SmPC) **[405]** Aspirin 75 mg tablets (UK SmPC) **[311]** NICE NG133

Full references p. 432–457